Where Do Negatives Come From?

Operational Closure, Group Completion, and Baselines

Abstract

Negative quantities are philosophically puzzling: they are indispensable in science and accounting, yet often seem metaphysically suspicious as “less-than-nothing.” This puzzle dissolves when we shift attention from ontology to practice. A negative is not an entity but a **functional role** within a **typed operational system** that supports **netting to a privileged neutral state**. Formally, genuine negatives occur exactly when a system’s states form (or are canonically extended to) an **abelian group** under its composition rule, so that each state x has an inverse $-x$ with $x + (-x) = 0$. This framework explains both the presence and the emergence of negatives: some domains contain inverses intrinsically (displacements, charge), while others acquire them when practices introduce deficits or obligations, modeled by **Grothendieck group completion** (e.g., inventory with backorders; double-entry bookkeeping). Baseline-relativity is captured by **torsor** structure, where inverses live in **differences** rather than absolute states. We distinguish directed deficits from mere absences and derive testable predictions about sign reasoning under different framings.

Keywords: negative numbers; abelian groups; Grothendieck group; group completion; torsors; affine space; measurement scales; accounting; cognition.

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1. The problem

Negatives often invite a pseudo-problem: *do negative numbers (or “negative quantities”) exist, or are they merely absences?* The everyday use of debt, elevation below sea level, electric charge, and displacement suggests that “negative” is not simple nonexistence. But metaphysical reification (“negative being”) can obscure more than it clarifies.

Thesis (quietist/pragmatist): “Existence” is the wrong predicate. The right question is structural and operational:

> **Does this practice support cancellation (netting) to a privileged neutral state?**

If yes, then “$-x$” is as real—i.e., as rule-governed and inference-supporting—as “$+x$” within that practice.

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2. The insight: operational closure under inversion

A minus sign on a page does not guarantee a genuine negative. Genuine negatives belong to systems where practitioners can **offset** states to reach a **neutral**.

2.1 The netting test

A domain instantiates directed negatives exactly when it supports a robust “offset” rule:

- For a state x, there is an opposed state $-x$
- Composing them yields neutrality: $x \oplus (-x) = 0$
- Offsetting behaves consistently (associativity/commutativity in the relevant practice)

This is the operational signature of additive inverses.

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3. Formal core

3.1 Typed operational systems

A **typed operational system** is a triple

$$ \mathcal S_T = (S_T,\; \oplus,\; 0) $$

where:

- S_T is the set of admissible states of type T
- $\oplus$ is a (practice-defined) composition rule
- 0 is a privileged neutral (baseline) state

> **Typing matters:** cancellation only makes sense within the same type ($-\$5$ cancels $+\$5$, not $+5$ apples).

3.2 Genuine negatives (ideal case): abelian groups

$\mathcal{S}_T$ supports genuine negatives iff $(S_T, \oplus, 0)$ is an **abelian group**.

Equivalently, for every $x \in S_T$ there exists $-x \in S_T$ such that:

$$ x \oplus (-x) = 0. $$

Define subtraction as addition with inverse: $x \ominus y := x \oplus (-y)$.

Interpretation: “Directedness” is not metaphysical force; it is the operational fact that states come with systematic counter-states enabling netting.

3.3 Relativity case: torsors (differences are invertible)

Many “absolute” state spaces have no canonical origin. What is operationally well-defined are **differences**.

A set A is a **torsor** over a group $(G, +, 0)$ if G acts freely and transitively on A. Intuitively: a torsor is a "group that forgot its identity element."

- Absolute positions form a torsor A over the displacement group $G \cong (\mathbb{R}^n, +)$
- For $a, b \in A$, the "difference" $b - a \in G$ is defined and is invertible
- Choosing a baseline $a_0 \in A$ coordinatizes points by $a \mapsto (a - a_0) \in G$

Moral: coordinate negatives (e.g., "left of the origin") are *baseline artifacts*; inversional negatives live in the difference group.

3.4 Emergence case: monoids and Grothendieck completion

Some practices begin with a commutative **monoid** $(M, \oplus, 0)$ (no inverses): cash-only tallies, inventories without backorders, counts of resources.

When the practice introduces obligations/deficits/backorders, it effectively extends M to a group: the **Grothendieck group completion** $G(M)$.

- Elements of $G(M)$ behave like formal differences (often written $[a, b]$)
- They enforce $[a, b] + [c, d] = [a \oplus c,\ b \oplus d]$
- $[a, b] = 0$ iff $a = b$ (in cancellative settings)

Interpretive gloss: negative inventory is the internalization of "would need to be supplied" into the state space.

Lemma (injectivity criterion; proof sketch)

If M is cancellative and commutative, the natural map $M \to G(M)$ is injective. (Cancellation prevents different "positive" states from collapsing under completion.)

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4. Algebraic taxonomy of domains (Table 1)

Domain	"States"	Composition	Structure	Where inverses live?	Minus means…
Cash-only tally	$\mathbb{N}$	$+$	Monoid	nowhere	not defined in-domain
Double-entry net balance	$\mathbb{Z}$	$+$	Abelian group	in states	inverse / cancellation
Inventory (no backorders)	$\mathbb{N}$	$+$	Monoid	nowhere	not licensed
Inventory (with backorders)	$G(\mathbb{N}) \cong \mathbb{Z}$	$+$	Group completion	in completed states	engineered inverse

Domain	"States"	Composition	Structure	Where inverses live?	Minus means...
Absolute position	A	(no canonical $+$)	Torsor over $(\mathbb{R}^n, +)$	in differences	coordinate choice
Displacement	$\mathbb{R}^n$	$+$	Abelian group	in states	inverse / cancellation

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5. Paradigm cases

5.1 Debt and credit (the canonical netting practice)

Model net balance as $x \in (\mathbb{Z}, +)$ or $(\mathbb{R}, +)$. A $+5$ credit and -5 debt satisfy:

$$ (+5)+(-5)=0. $$

Multi-agent transfer naturally yields invariants: $(x_A, x_B) \mapsto (x_A - 5,\ x_B + 5)$ preserves $x_A + x_B$ in a closed system.

5.2 Inventory and backorders (emergence via completion)

- Without backorders: inventory is a monoid $(\mathbb{N}, +)$
- Allowing backorders adds deficits, producing a group-like bookkeeping domain isomorphic to $\mathbb{Z}$

This is a clean illustration that negatives can be **engineered** into a practice when netting is operationally required.

5.3 Displacement vs absolute position (torsor vs group)

- Absolute positions (without a distinguished origin) form a torsor A
- Displacements form a group $(\mathbb{R}^n, +)$
- "Negative position" is not intrinsic; it arises only after choosing coordinates (a baseline point + axis)

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6. Demarcation from mere absence

"Mere absences" (e.g., "the absence of unicorns") typically fail the netting test:

- no agreed composition $\oplus$
- no privileged neutral 0
- no operationally meaningful inverse $-x$

By contrast, directed deficits (debt, backorders, charges) are embedded in systems with explicit netting rules, conservation constraints, and normative or interactional governance.

7. Not every minus sign indicates a genuine negative (measurement sidebar)

Many measurement systems are **affine** (interval scales): the zero point is conventional, and meaningful structure lives in **differences**.

- Celsius readings allow negatives because the origin is chosen by convention.
- What is operationally stable is the **temperature difference** ΔT, which supports additive inverses in $(\mathbb{R}, +)$.

This aligns with measurement theory: interval scales are invariant under positive affine transformations $x \mapsto ax + b$ (Stevens).

Moral: genuine inversional negativity belongs to the operational algebra of the relevant quantity (often differences), not to the glyph "−."

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8. Philosophical payoff: dissolving "Do negatives exist?"

This framework resolves a longstanding pseudo-problem.

- Asking whether "$-x$ exists" abstracts away from the practice that gives "$-x$" content.
- Within a practice whose state space is (or is completed to) a group, "$-x$" is a legitimate, indispensable role.

> **Negatives are not discovered in the world; they are engineered into practices that require netting.**

This is quietist about metaphysical "negative being," but not deflationary about mathematics: it explains why negatives are indispensable, stable, and transferable across domains—because many practices share the same algebraic requirements.

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9. Predictions and validation

9.1 Conceptual audit (fast)

Survey domains and classify them as group / torsor / monoid / completion, recording:

- states, composition, baseline, inverse availability
- whether "minus" is inversional (group) or coordinate/affine (torsor chart)

9.2 Formal validation (fast)

Prove representation results of the form:

- If a practice supports netting with associativity/commutativity and inverses, it is modeled by an abelian group.
- If a practice is monoidal but cancellative, group completion is canonical and explains deficit-states.
- If a practice admits only differences, torsor structure captures baseline relativity.

9.3 Cognitive prediction (cheap experiment)

People should reason more accurately and quickly about negatives when framed in netting practices (debt/credit, displacement) than when framed as bare symbol manipulation or "mere absence."

A simple design:

- Same algebraic task (e.g., solve $x - 5 = 2$)
- Framed as: (A) debt, (B) displacement, (C) pure symbols, (D) "absence/hole" metaphor
- Measure accuracy, time, and systematic sign errors

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10. Optional enrichments (when you *do* get vectors)

Vector spaces (or ordered groups, cones, etc.) are **enrichments**, not the foundation.

- If the domain supports meaningful scalar action (e.g., "triple the debt"), then the abelian group becomes a $\mathbb{Z}$-module (often extendable to $\mathbb{R}$-vector space for continuous quantities).
- If the domain supports a partial order (e.g., 'at least as good as'), you may add ordered-group structure.

These additions are domain-dependent; the core litmus test remains operational inversion (or its canonical emergence via completion).

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